

BYRON CHIN

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Georgia Tech School of Mathematics
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ACADEMIC POSITIONS

Georgia Institute of Technology

2026-present

Hale Visiting Assistant Professor, School of Mathematics
Postdoctoral Fellow, Algorithms and Randomness Center

EDUCATION

Massachusetts Institute of Technology

2021-2026

Ph.D. Mathematics, advised by Elchanan Mossel

Princeton University

2017-2021

A.B. Mathematics, with high honors

RESEARCH INTERESTS

Discrete Probability, High-dimensional Statistics, Combinatorics

↔ community detection, random graphs, phase transitions, large deviations, Markov chains

AWARDS

- NSF Graduate Research Fellowship 2021-2026
- MIT Akamai Presidential Fellowship 2021
- Best Independent Work – Princeton Statistics and Machine Learning 2021
- Princeton Shapiro Prize for Academic Excellence 2019

PAPERS

11. Extremal subtrees of critical beta-splitting trees.
with Anna Brandenberger, Elchanan Mossel.
submitted.
10. The random stable roommates problem typically has no solution.
with Marcus Michelen.
submitted. [arXiv:2601.07612](https://arxiv.org/abs/2601.07612)
9. Asymptotics for the harmonic descent chain and applications to critical beta-splitting trees.
with Anna Brandenberger, Elchanan Mossel.
under revision at *Annales Henri Lebesgue*. [arXiv:2505.24821](https://arxiv.org/abs/2505.24821)
8. Stochastic block models with many communities and the Kesten–Stigum bound.
with Elchanan Mossel, Youngtak Sohn, Alexander S. Wein.
under revision at *SIAM Journal on Computing*.
conference version in *Conference on Learning Theory (COLT 2025)*. [arXiv:2503.03047](https://arxiv.org/abs/2503.03047)
7. Exact threshold and limiting distribution for non-linear Hamilton cycles.
Combinatorics, Probability and Computing, to appear. [arXiv:2411.13452](https://arxiv.org/abs/2411.13452)
6. Matching algorithms in sparse stochastic block models.
with Anna Brandenberger, Nathan Sheffield, Divya Shyamal.
Analysis of Algorithms (AofA 2024). [arXiv:2403.02140](https://arxiv.org/abs/2403.02140)

5. Structure of lower tails in sparse random graphs.
Random Structures & Algorithms **67** (2025), e70028. [arXiv:2312.12673](#)
4. The power of an adversary in Glauber dynamics.
with Ankur Moitra, Elchanan Mossel, Colin Sandon.
Conference on Learning Theory (COLT 2024). [arXiv:2302.10841](#)
3. When will (game) wars end?
with Manan Bhatia, Nitya Mani, Elchanan Mossel.
The American Mathematical Monthly **133**(1) (2026), 35-46. [arXiv:2302.03535](#)
2. Optimal reconstruction of general sparse stochastic block models.
with Allan Sly.
Annales de l'Institut Henri Poincaré, to appear. [arXiv:2111.00697](#)
Princeton SML Best Independent Work.
1. Optimal recovery of block models with q communities.
with Allan Sly.
[arXiv:2010.10672](#)

TALKS

Invited

- JMM Special Session on Discrete Random Structures
TBD *January 2027*
- Georgia Tech Combinatorics Seminar *February 2026*
Exact threshold for nonlinear Hamilton cycles
- Georgia Tech Algorithms and Randomness Colloquium *February 2026*
Solvability of the random stable roommates problem
- Northwestern Probability Seminar *October 2025*
A few central limit theorems for critical beta-splitting trees
- Rocky Mountain Summer Workshop on Algorithms, Probability and Combinatorics *June 2025*
Stochastic block models with many communities and the Kesten–Stigum bound
- Richard P. Stanley Seminar in Combinatorics *December 2024*
Exact threshold for nonlinear Hamilton cycles
- MIT Probability Seminar *May 2024*
Structure of lower tails in sparse random graphs

Conference

- Northeast Probability Seminar 2025 *November 2025*
Stochastic block models with many communities and the Kesten–Stigum bound
- COLT 2025 *July 2025*
Stochastic block models with many communities and the Kesten–Stigum bound
- COLT 2024 *July 2024*
The power of an adversary in Glauber dynamics
- AofA 2024 *June 2024*
Matching algorithms in the sparse stochastic block model

Other

- Berkeley Student Probability Seminar *February 2025*
The power of an adversary in Glauber dynamics
- MIT Probability Group Meeting *May 2024*
Structure of lower tails in sparse random graphs

- MIT SPAMS *April 2022*
Reconstruction of the stochastic block model
- MIT Probability Group Meeting *April 2022*
Reconstruction of the stochastic block model

TEACHING

- 1553 Introduction to Linear Algebra (Georgia Tech), Instructor *Fall 2026*
- 18.600 Probability and Random Variables (MIT), TA/Recitation Leader *Fall 2024*

MENTORING/ORGANIZATION

- Co-organizer of MIT SPAMS graduate student seminar *2024*
- MIT Summer Program in Undergraduate Research (SPUR) mentor *2023*
- Princeton University Math Competition officer *2019*
- Co-founder of West Windsor Math Advancement Center *2016*

REVIEWING

Journals

- Probability Theory and Related Fields

Conferences

- COLT, ITCS